

# LCMV Infection Models (WE, Clone 13, Docile, Armstrong)

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**Protocol ID:** MUS-INF-LCMV

**Version:** v1.1

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## Purpose

This protocol describes standardized mouse infection procedures using lymphocytic choriomeningitis virus (LCMV), with primary emphasis on **LCMV-WE** and **LCMV-Clone 13**, and additional reference to **LCMV-Docile** and **LCMV-Armstrong**.

The protocol defines virus handling, dilution, injection parameters, infection doses, and sampling windows, with specific annotation of **liver NLC (NK-like cell) dynamics** to guide optimal experimental design.

## Critical notes (read before starting)

- All LCMV work must be performed under **approved animal and biosafety protocols**.
- Virus stocks must be **titered, aliquoted, and freeze-thaw limited**.
- Infection dose and route strongly influence antigen load and immune kinetics.
- **NLC dynamics differ markedly between WE and Clone 13**, requiring strain-specific sampling strategies.
- Late timepoints (> day 7) in Clone 13 infection are **suboptimal for NLC-focused analyses**.

## Approximate timing

- Virus thawing and dilution: **5–10 min**
- Intravenous injection: **~1–2 min per mouse**
- Acute infection window: **days 0–15**
- Chronic infection window: **days 0–30+**

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## Procedure

### Virus stock handling and preparation

LCMV-WE, LCMV-Clone 13, LCMV-Docile, and LCMV-Armstrong virus stocks are propagated and titrated as previously described in the literature and internal Abdullah Lab SOPs.

- Virus stocks are stored at **−80 °C** in single-use aliquots.
- Repeated freeze–thaw cycles must be avoided.

### Thawing

- Virus aliquots are **rapidly thawed immediately before use**.
- Thawing is performed in a **37 °C water bath** until the aliquot is just thawed.
- Aliquots are removed promptly from the water bath and placed **immediately on ice**.
- Virus is kept on ice during dilution and injection.
- Repeated freeze–thaw cycles must be avoided.

### Dilution buffer

- Virus stocks are diluted in **sterile, endotoxin-free PBS**.
- No serum or supplements are added unless explicitly required by internal SOP.
- Dilutions are prepared **fresh on the day of infection**.

### Infection doses and routes

Mice are infected **intravenously (i.v.)** via the lateral tail vein using the following Abdullah Lab standard doses:

- **LCMV-Clone 13:  $2 \times 10^5$  plaque-forming units (PFU)**
- **LCMV-WE:  $2 \times 10^5$  plaque-forming units (PFU)**
- **LCMV-Armstrong:  $2 \times 10^5$  plaque-forming units (PFU)**
- **LCMV-Docile:  $2 \times 10^6$  plaque-forming units (PFU)**

#### **i** Note

**LCMV-Armstrong — route note:** The standard Abdullah Lab dose for Armstrong is  **$2 \times 10^5$  PFU i.v.** (Anna’s protocol). The broader literature typically uses  **$1 \times 10^5 - 2 \times 10^5$  PFU administered intraperitoneally (i.p.)** for acute infection and memory studies. The i.p. route is the most common in published Armstrong protocols. The i.v. route used here differs from the literature standard — this should be noted when comparing outcomes across studies.

## Injection parameters

- **Injection volume: 200  $\mu$ L per mouse**
- **Syringe:** insulin syringe
- **Needle gauge: 27G–30G**
- Injections are performed slowly to ensure proper intravenous delivery.
- Successful i.v. injection is confirmed by lack of resistance and absence of subcutaneous bleb formation.

## Post-infection monitoring

- Body weight and general health are monitored daily during the first week post infection.
- Monitoring frequency is adjusted according to infection severity and institutional guidelines.
- Humane endpoints are applied as defined in approved protocols.

## Infection dose ranges, rationale, and expected outcomes

| Virus strain     | Dose range<br>(PFU,<br>literature)   | Route<br>(literature)             | Rationale for<br>low vs high<br>dose                                              | Expected<br>outcome                                                                 | Abdullah Lab<br>standard                                     |
|------------------|--------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Armstrong</b> | $\sim 1 \times 10^4 - 2 \times 10^4$ | <b>i.p.</b> (literature standard) | Low–moderate dose to induce acute infection with rapid immune control             | Acute infection; strong effector CD8 response; viral clearance and memory formation | <b><math>2 \times 10^4</math> PFU i.v.</b> (Anna’s protocol) |
| <b>WE</b>        | <b><math>2 \times 10^4</math></b>    | i.v.                              |                                                                                   |                                                                                     | <b>Yes</b>                                                   |
|                  | $\sim 1 \times 10^2 - 1 \times 10^3$ | i.v.                              | Minimal acute infection                                                           | Rapid clearance; limited systemic and hepatic involvement                           | –                                                            |
|                  | <b><math>2 \times 10^4</math></b>    | i.v.                              | Intermediate dose to ensure robust systemic infection without chronic persistence | Strong liver immune response; sustained NLC accumulation                            | <b>Yes</b>                                                   |
|                  | $\sim 1 \times 10^4$                 | i.v.                              | High-dose systemic challenge used in comparative studies                          | Severe acute disease; heightened inflammation                                       | No                                                           |

| Virus strain    | Dose range<br>(PFU,<br>literature) | Route<br>(literature) | Rationale for<br>low vs high<br>dose                                      | Expected<br>outcome                                              | Abdullah Lab<br>standard |
|-----------------|------------------------------------|-----------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------|
| <b>Clone 13</b> | <b><math>2 \times 10</math></b>    | i.v.                  | High antigen<br>load to enforce<br>chronic<br>infection and<br>exhaustion | Persistent<br>infection; CD8<br>exhaustion<br>program            | <b>Yes</b>               |
| <b>Docile</b>   | $\sim 1 \times 10$                 | i.v./i.p.             | Moderate dose<br>allowing<br>partial control                              | Semi-chronic<br>infection;<br>intermediate<br>exhaustion         | –                        |
|                 | <b><math>2 \times 10</math></b>    | i.v.                  | High-dose<br>infection to<br>drive persistent<br>antigen<br>exposure      | Chronic<br>infection;<br>exhaustion-<br>prone immune<br>response | <b>Yes</b>               |

## Expected NLC dynamics in the liver

### LCMV-WE ( $2 \times 10$ PFU i.v.)

- NLCs accumulate prominently in the liver.
- Peak frequency occurs at approximately **day 7 post infection**.
- Elevated NLC levels are **maintained through at least day 30 post infection**.
- This model supports **both early and long-term NLC analyses**.

### LCMV-Clone 13 ( $2 \times 10$ PFU i.v.)

- NLCs peak **early**, around **day 5 post infection**.
- A **sharp contraction occurs by ~day 7**, followed by gradual decline through day 30.
- The effective experimental window for NLC analysis is **day 4–7 post infection**.
- Sampling outside this window will substantially underestimate NLC abundance.

## Experimental design guidance

- For studies requiring **broad or late NLC sampling**, LCMV-WE is preferred.
- For studies focused on **early inflammatory or antigen-rich environments**, Clone 13 is appropriate but requires tight temporal control.
- Direct comparisons between WE and Clone 13 must account for **non-overlapping NLC kinetics**.

## Buffers used

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

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Warning: package 'readr' was built under R version 4.5.2

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Warning: package 'lubridate' was built under R version 4.5.2

None

## Safety (brief)

- All animal procedures must comply with approved animal licences and institutional regulations.
- Wear appropriate PPE (lab coat, gloves, eye protection) when handling animals, tissues, and reagents.
- Handle sharps (needles, scalpels) with care and dispose of immediately in approved sharps containers.
- Perform virus handling, injections, and tissue processing in accordance with institutional biosafety guidelines.
- Dispose of viral waste and contaminated materials according to local biosafety regulations.

## Version history

| Version | Date       | Author         | Change summary                                                                                                                                                                             |
|---------|------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.0    | 2026-01-09 | Dillon Corvino | Initial LCMV infection protocol (WE, Clone 13, Docile, Armstrong)                                                                                                                          |
| v1.1    | 2026-04-13 | Dillon Corvino | Added Armstrong as Abdullah Lab standard ( $2 \times 10$ PFU i.v., Anna's protocol); added route note clarifying i.p. vs i.v. difference from literature; added Route column to dose table |